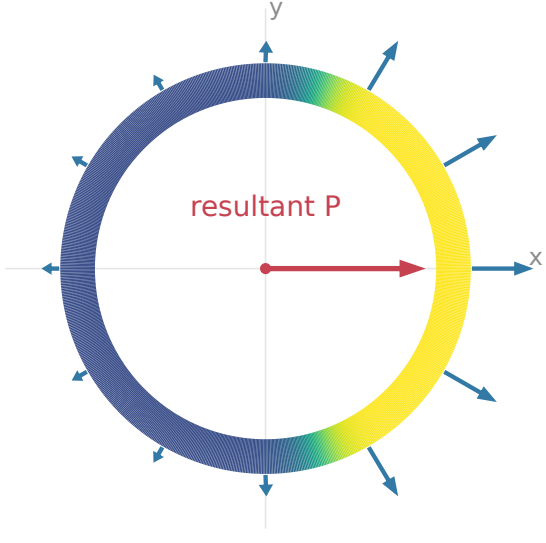
Two activity peaks can leave a weak resultant

Activity u is real; x + iy stores its weighted normal's two coordinates.

$$M_1 = \sum_j u_j \Delta s_j (\cos\theta_j + i\sin\theta_j), \qquad M_0 = \sum_j u_j \Delta s_j$$

$P = M_1/M_1(0)$ rotates and scales the initial resultant to 1.

A Initial basal activity



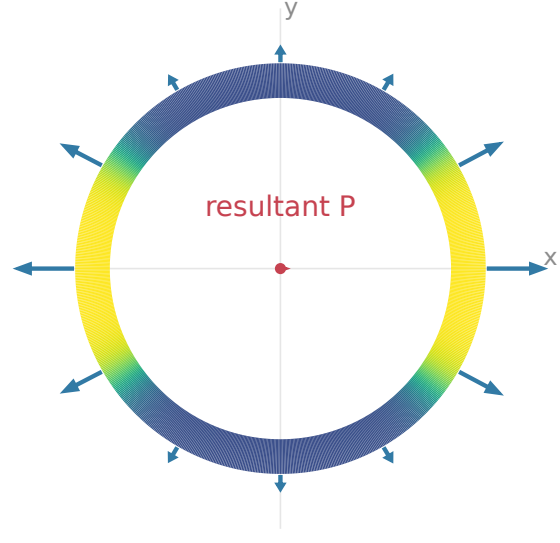
Before the antipodal cue

Total activity $M_0/M_0(0) = 1.0000$

Resultant magnitude $R = |P| = 1.0000$

Initial normalization sets P(0) = 1.

B Retained endpoint: opposing peaks



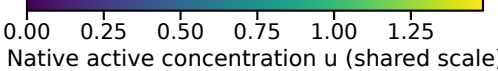
1000 cue units + 13,000 cue-free units

Total activity $M_0/M_0(0) = 0.9975$

Resultant magnitude $R = |P| = 0.0399$

Heading is not interpreted when R < 0.1.

Blue arrows: exact sums of native cell contributions in 30° sectors.



All arrows use one vector scale.
Holmes 3 · fixed circumference C = 40.